A bounded L^1 martingale with no admissible weak square-function decomposition

Autonomous counterexample packet for arXiv:math/0111264

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Abstract

Problem 6.6 of Randrianantoanina asks for a weak-L^1 row/column square-function estimate for every bounded L^1 martingale, with the decomposition pieces required to belong to the strong Hardy spaces $\mathcal{H}_C^1$ and $\mathcal{H}_R^1$. The statement as printed is false already in a commutative probability space. We construct a positive, uniformly integrable martingale x with $\sup_n \|x_n\|_1 < \infty$ but $\|S(x)\|_1 = \infty$. In the commutative case both row and column Hardy spaces equal H^1, so any admissible decomposition would force $x \in H^1$.

1 The source problem

Problem 6.6 asks whether there is a constant K such that every bounded L^1 martingale x satisfies

$$\inf_{\substack{x=y+z\\ y\in\mathcal{H}_C^1,\ z\in\mathcal{H}_R^1}} \left(\|dy\|_{L^{1,\infty}(\mathcal{M};\ell_C^2)} + \|dz\|_{L^{1,\infty}(\mathcal{M};\ell_R^2)}\right) \le K\|x\|_1.$$

Here is the actual statement from source PDF page 29:

> **Problem 6.6.** Does there exist a constant K such that for every bounded L^1-martingale x
>
> $$\inf\left\{\|dy\|_{L^{1,\infty}(\mathcal{M};l_C^2)} + \|dz\|_{L^{1,\infty}(\mathcal{M};l_R^2)} :\ x = y+z,\ y \in \mathcal{H}_C^1(\mathcal{M}),\ z \in \mathcal{H}_R^1(\mathcal{M})\right\} \le K\,\|x\|_1?$$

The strong memberships $y \in \mathcal{H}_C^1$ and $z \in \mathcal{H}_R^1$ are decisive. We show that for one bounded L^1 martingale there is no such pair at all.

2 The martingale

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a nonatomic probability space partitioned into pairwise disjoint sets A_k, $k \ge 1$, with $\mathbb{P}(A_k) = p_k = 2^{-k}$. Inside each A_k choose nested sets

$$A_k = I_{k,0} \supset I_{k,1} \supset \cdots \supset I_{k,k}, \qquad \mathbb{P}(I_{k,j}) = p_k 2^{-j},$$

and put $J_{k,j} = I_{k,j-1} \setminus I_{k,j}$. Thus $\mathbb{P}(J_{k,j}) = p_k 2^{-j}$.

Set $a_k = k^{-2}$ and $b_k = a_k/p_k$. Define the nonnegative terminal variable

$$f = \sum_{k\ge1} b_k 2^k \mathbf{1}_{I_{k,k}}.$$

Its supports are disjoint, and hence

$$\|f\|_1 = \sum_{k\geq 1} b_k 2^k \mathbb{P}(I_{k,k}) = \sum_{k\geq 1} a_k < \infty.$$

Let $\mathcal{F}_1 = \sigma(A_k : k \geq 1)$. For $n \geq 1$, obtain $\mathcal{F}_{n+1}$ by revealing, in every block A_k with $k \geq n$, the next split $I_{k,n} \subset I_{k,n-1}$; blocks whose depth has been exhausted remain unchanged. Equivalently, $\mathcal{F}_{n+1}$ is generated by all A_k and all $I_{k,j}$ with $1 \leq j \leq \min\{n, k\}$. Put

$$x_n = \mathbb{E}(f \mid \mathcal{F}_n), \qquad n \geq 1.$$

Then x is a positive, uniformly integrable martingale and

$$\sup_n \|x_n\|_1 = \|f\|_1 = \sum_{k\geq 1} k^{-2}.$$

On A_k its values are especially simple:

$$x_1 = b_k \mathbf{1}_{A_k}, \qquad x_{j+1} = b_k 2^j \mathbf{1}_{I_{k,j}} \quad (1 \leq j \leq k),$$

and it is constant thereafter. With the convention $x_0 = 0$, the local difference at level j is therefore

$$dx_{j+1} = \begin{cases} b_k 2^{j-1}, & \text{on } I_{k,j}, \\ -b_k 2^{j-1}, & \text{on } J_{k,j}, \\ 0, & \text{on } A_k \setminus I_{k,j-1}. \end{cases}$$

3 Divergence of the square function

Theorem 1. *The martingale x is bounded in L^1, but its ordinary square function*

$$S(x) = \left(\sum_{n\geq 1} |dx_n|^2 \right)^{1/2}$$

does not belong to L^1.

Proof. Fix $k \geq 1$ and $1 \leq j \leq k$. On $J_{k,j}$ the square function dominates the absolute value of the $(j+1)$st difference, so

$$S(x) \geq b_k 2^{j-1} \quad \text{on } J_{k,j}.$$

Consequently,

$$\int_{J_{k,j}} S(x)\, d\mathbb{P} \geq b_k 2^{j-1} p_k 2^{-j} = \frac{a_k}{2}.$$

The sets $J_{k,j}$ are mutually disjoint over all (k, j). Summing gives

$$\|S(x)\|_1 \geq \frac{1}{2} \sum_{k\geq 1} \sum_{j=1}^{k} a_k = \frac{1}{2} \sum_{k\geq 1} \frac{1}{k} = \infty.$$

$\square$

4 Negative answer to Problem 6.6

Theorem 2. *There is no universal constant in Problem 6.6. Indeed, for the martingale x above, the set of decompositions over which the displayed infimum is taken is empty.*

Proof. View the probability space as the abelian finite von Neumann algebra $\mathcal{M} = L^\infty(\Omega)$. In an abelian algebra the column and row square functions coincide with S. Suppose that $x = y + z$ with $y \in \mathcal{H}^1_C$ and $z \in \mathcal{H}^1_R$. Pointwise Minkowski's inequality in ℓ^2 gives

$$S(x) = \left(\sum_n |dy_n + dz_n|^2 \right)^{1/2} \leq S(y) + S(z).$$

The right-hand side is integrable by the two strong Hardy memberships. It follows that $S(x) \in L^1$, contrary to Theorem 1. No admissible pair (y, z) exists, so the infimum is $+\infty$ and cannot be bounded by $K\|x\|_1$. $\square$

5 Relation to later dense-subcase results

Later papers proved weak row/column square-function decompositions for L^2 martingales with an L^1 right-hand side. That does not contradict the example: the terminal variable above need not be in L^2, and the strong Hardy-membership requirement in the universal L^1 formulation already makes the admissible class empty. A viable endpoint question would have to enlarge the decomposition class to a weak Hardy completion (or restrict the input class), rather than require both pieces to lie in strong H^1.

References

[1] N. Randrianantoanina, *Non-commutative martingale transforms*, arXiv:math/0111264, Problem 6.6, PDF page 29.